Topic: Smart Waste Management System

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Abstract:

This report evaluates a dual-approach smart waste management system. The first approach uses an Arduino-based sensor system for real-time bin fill monitoring, and the second uses deep learning for automated waste classification. Evaluation shows that a Random Forest model predicts bin status with 89% accuracy, while a custom-built CNN classifies waste images with 70% accuracy. The report details hardware configurations, model architectures, performance metrics, and practical implementation workflows.

1.Introduction:

This report describes a smart waste management system that uses two different approaches to identify and sort waste:

1. **A sensor-based system**: Uses an Arduino with ultrasonic and infrared sensors to measure how full a bin is.
2. **An image classification system:** Uses deep learning models to identify different types of waste (cardboard, glass, metal, paper, plastic, trash).

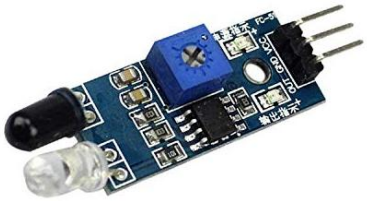
The goal is to create an automated system that detects bin fullness and identifies waste types for better sorting and collection management.

2. Part 1: Sensor-Based Bin Monitoring (Arduino):

2.1 Hardware Components

- **Ultrasonic sensor** (trigPin 9, echoPin 10): Measures distance to the waste surface.

- **Infrared (IR) sensor** (irPin 7): Detects if an object is present.



- **Arduino board**: Processes sensor data.

2.2 Functional Logic

The system averages five distance measurements for reliability. The fill level percentage is calculated using the following formula:

$$\text{Fill Level} = \left(\frac{\text{Bin Height} - \text{Distance}}{\text{Bin Height}} \right) \times 100$$

Example: If a bin is 30 cm tall and the waste is 6 cm from the top, the fill level is 80%.

2.3 Sensor Readings and Output

The system outputs the following data every 5 seconds:

- Distance in centimetres.
- Fill level percentage.
- Alert message for fill levels exceeding 80%.
- IR sensor status.

Sample Output:

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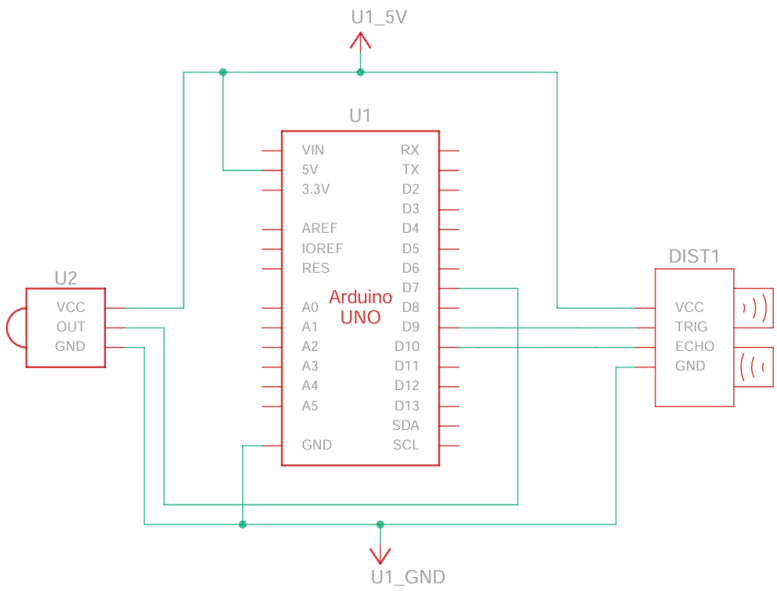
Distance: 4.98 cm
Fill Level: 83.40 %
ALERT: Bin Full!
IR: Object Detected
```

<u>Circuit Diagram:</u>



3. Part 2: Waste Classification Using Machine Learning:

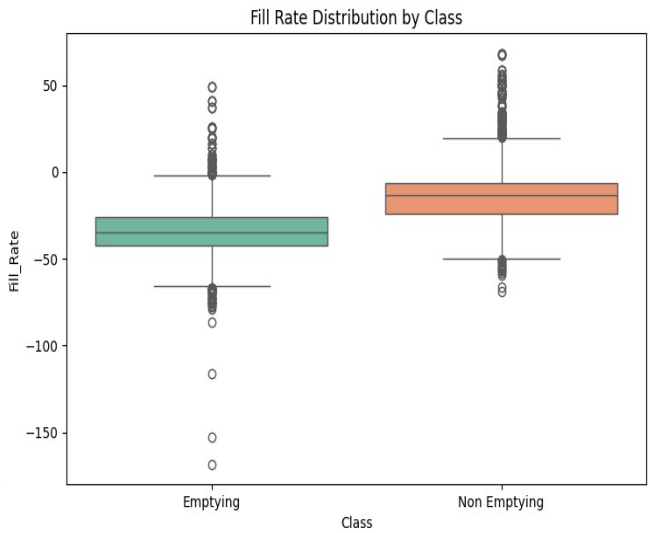
3.1 Approach A: Random Forest Classifier (Sensor Data)

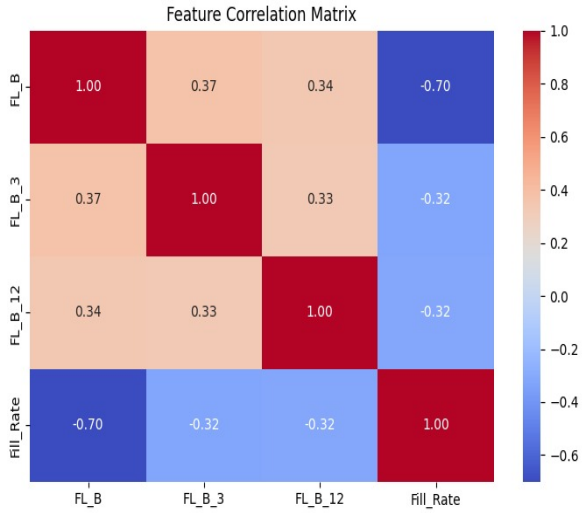
This model predicts if a bin needs emptying based on sensor data.

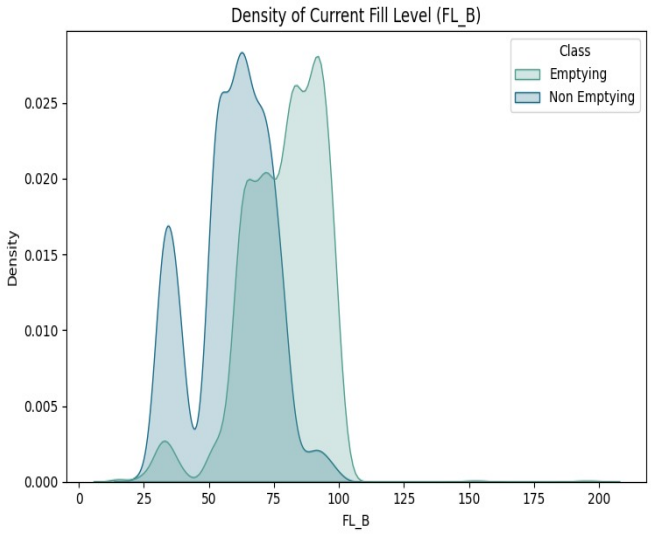
Input Features:

- **FL_B**: Current fill level reading.
- **FL_B_3**: Fill level from 3 minutes ago.
- **FL_B_12**: Fill level from 12 minutes ago.
- **Filtrate**: Rate of filling $\mathrm{FL}_A - \mathrm{FL}_B$ (calculated as).

Feature Extraction-:







Model Performance:

```
Classification Report:
              precision    recall  f1-score   support

    Emptying       0.90      0.87      0.88       455
Non Emptying       0.88      0.90      0.89       473

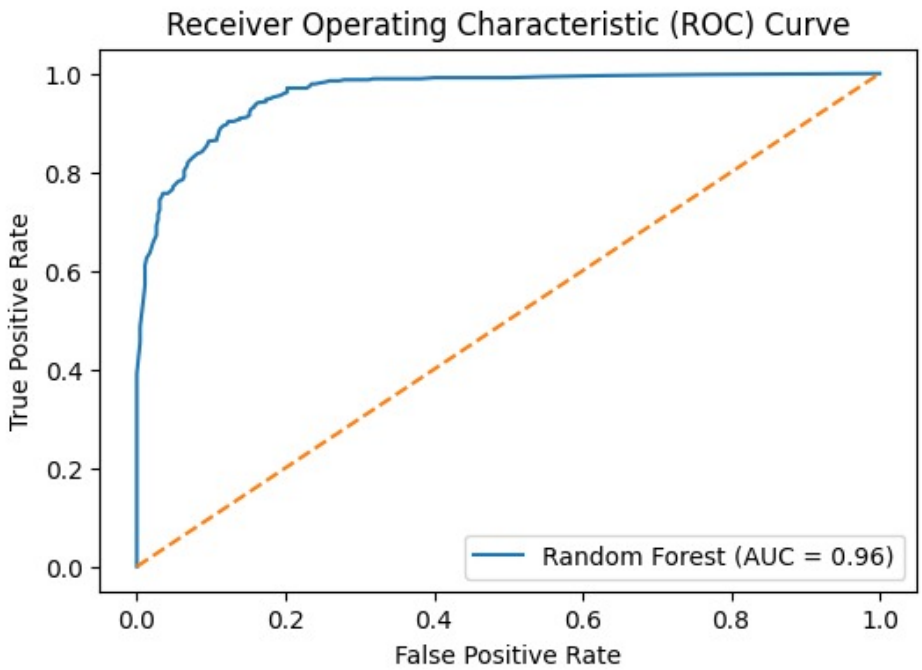
    accuracy                           0.89       928
   macro avg       0.89      0.89      0.89       928
weighted avg       0.89      0.89      0.89       928


Confusion Matrix:
                     Predicted Emptying  Predicted Non Emptying
Actual Emptying                     397                      58
Actual Non Emptying                  46                     427

ROC-AUC Score: 0.9605
```

The model correctly identified 397 bins requiring emptying and 427 that did not. It recorded 58 false alarms and 46 missed detections.

AUC-ROC Curve:



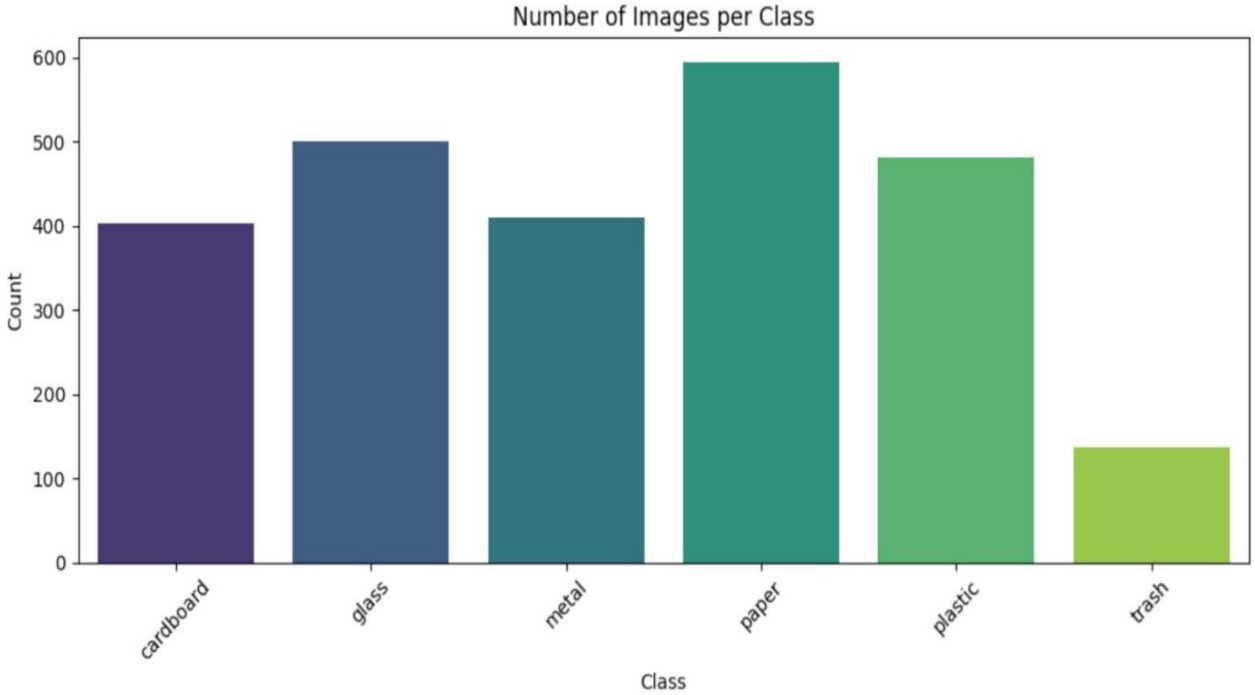
3.2 Approach B: Deep Learning Models (Image Based)

Dataset Information:

- **Total Images:** 2,527 (Resized to 224 * 224 pixels).
- **Classes:** Cardboard (403), Glass (501), Metal (410), Paper (594), Plastic (482), Trash (137).

```
EXPLORATORY DATA ANALYSIS

Dataset Statistics:
    Class  Count Sample Shape (W, H) Mode
cardboard    403           512 x 384  RGB
    glass    501           512 x 384  RGB
    metal    410           512 x 384  RGB
    paper    594           512 x 384  RGB
  plastic    482           512 x 384  RGB
    trash    137           512 x 384  RGB
```



3.2.1 Custom CNN (Built from Scratch)

Architecture: 4 Conv2D layers (32,64,128,256 filters), Max Pooling, Flatten, Dropout (0.5), Dense (256), and Dense (6 classes).

Results:

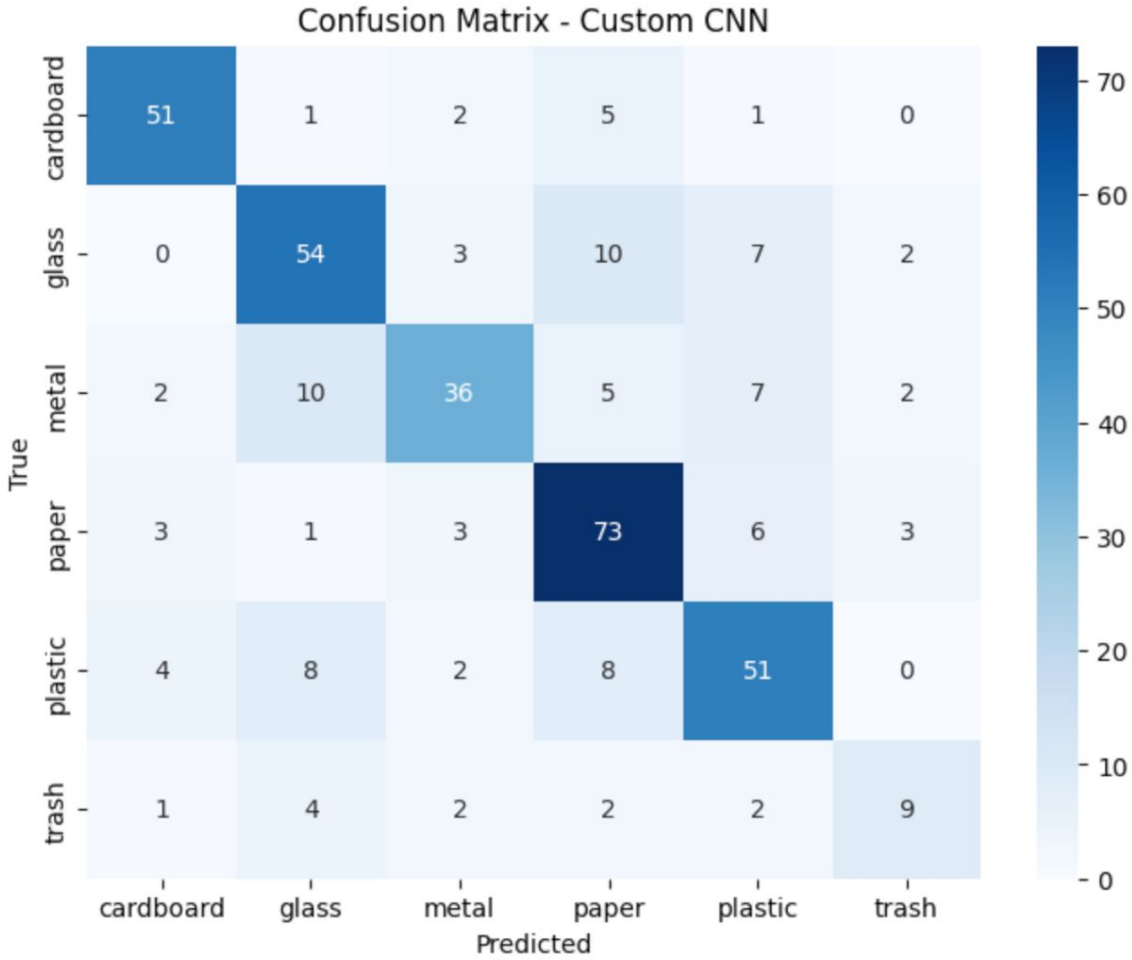
- **Test Accuracy:** 70.0%
- **Weighted F1 Score:** 0.70

```
Custom CNN - Test Accuracy: 0.7211, Weighted F1: 0.7179

Classification Report:
              precision    recall  f1-score   support

   cardboard       0.84      0.85      0.84        60
       glass       0.69      0.71      0.70        76
       metal       0.75      0.58      0.65        62
       paper       0.71      0.82      0.76        89
     plastic       0.69      0.70      0.69        73
       trash       0.56      0.45      0.50        20

    accuracy                           0.72       380
   macro avg       0.71      0.69      0.69       380
weighted avg       0.72      0.72      0.72       380
```



3.2.2 MobileNetV2 (Transfer Learning)

- **Test Accuracy:** 28.2%
- **Weighted F1 Score:** 0.21
- **Observation:** The model failed to generalize, predicting "paper" for over 60% of test samples.

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MobileNetV2 - Test Accuracy: 0.2763, Weighted F1: 0.2028

Classification Report:
              precision    recall  f1-score   support

   cardboard       1.00      0.02      0.03        60
       glass       0.26      0.16      0.20        76
       metal       0.00      0.00      0.00        62
       paper       0.28      0.61      0.38        89
     plastic       0.27      0.52      0.36        73
       trash       0.00      0.00      0.00        20

    accuracy                           0.28       380
   macro avg       0.30      0.22      0.16       380
weighted avg       0.33      0.28      0.20       380
```

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2026-04-06 22:12:57.196114: I tensorflow/core/framework/local_rendezvous.cc:407] Lo
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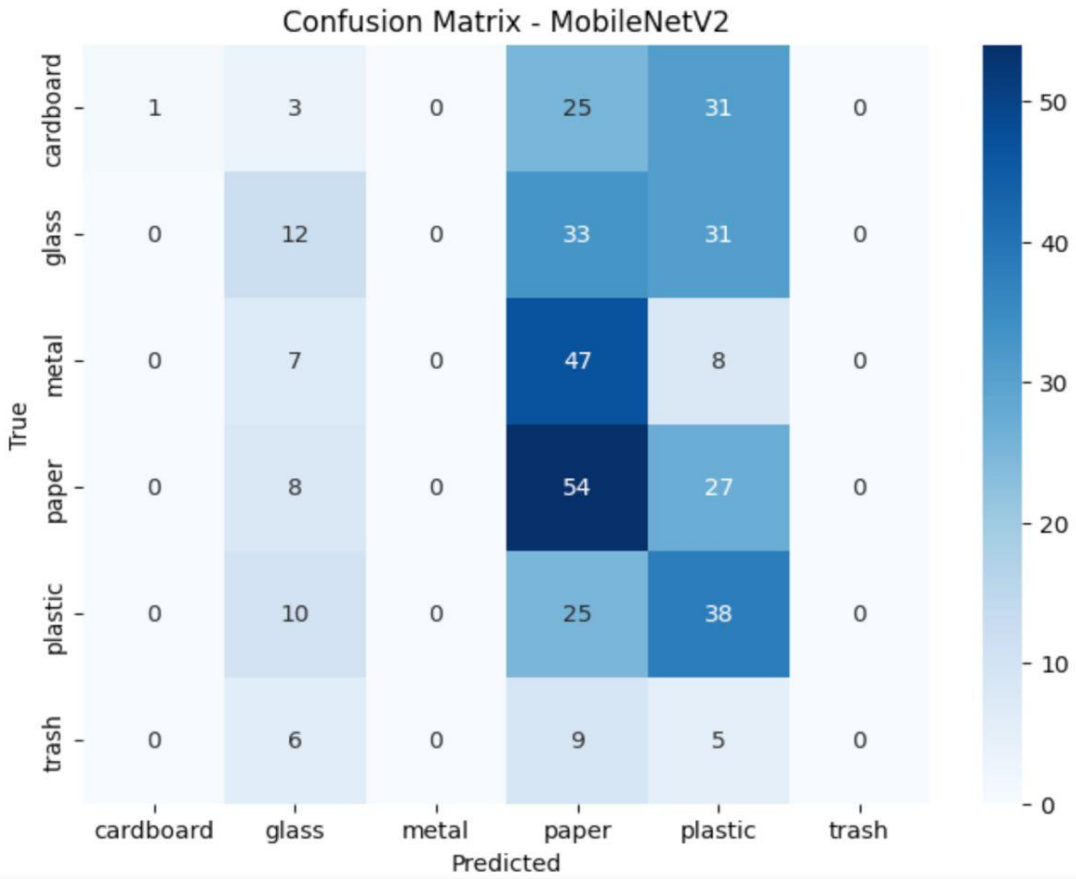

3.2.3 EfficientNetB0 (Transfer Learning)

- **Test Accuracy:** 23.4%
- **Weighted F1 Score:** 0.09
- **Observation:** Complete model failure; all samples were classified as "paper" due to dataset imbalance and poor feature transfer.

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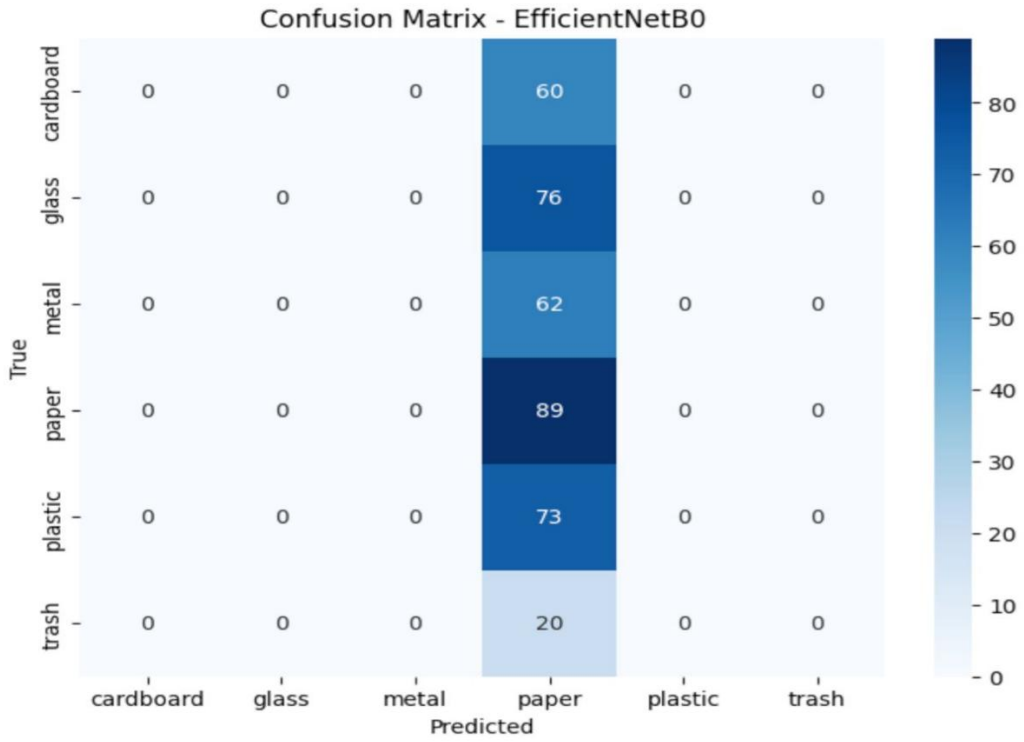
EfficientNetB0 - Test Accuracy: 0.2342, Weighted F1: 0.0889

Classification Report:
              precision    recall  f1-score   support

   cardboard       0.00      0.00      0.00        60
       glass       0.00      0.00      0.00        76
       metal       0.00      0.00      0.00        62
       paper       0.23      1.00      0.38        89
     plastic       0.00      0.00      0.00        73
       trash       0.00      0.00      0.00        20

    accuracy                           0.23       380
   macro avg       0.04      0.17      0.06       380
weighted avg       0.05      0.23      0.09       380
```



4. Comparison of All Models:

Model	Test Accuracy	Weighted F1	Best for...
Custom CNN	70.0%	0.70	General classification
MobileNetV2	28.2%	0.21	Not recommended
EfficientNetB0	23.4%	0.09	Not recommended

```
MODEL COMPARISON
        Model  Test Accuracy  Weighted F1 Score
    Custom CNN       0.721053           0.717866
   MobileNetV2       0.276316           0.202763
EfficientNetB0       0.234211           0.088890

🏆 BEST MODEL: Custom CNN (Test Accuracy = 0.7211)
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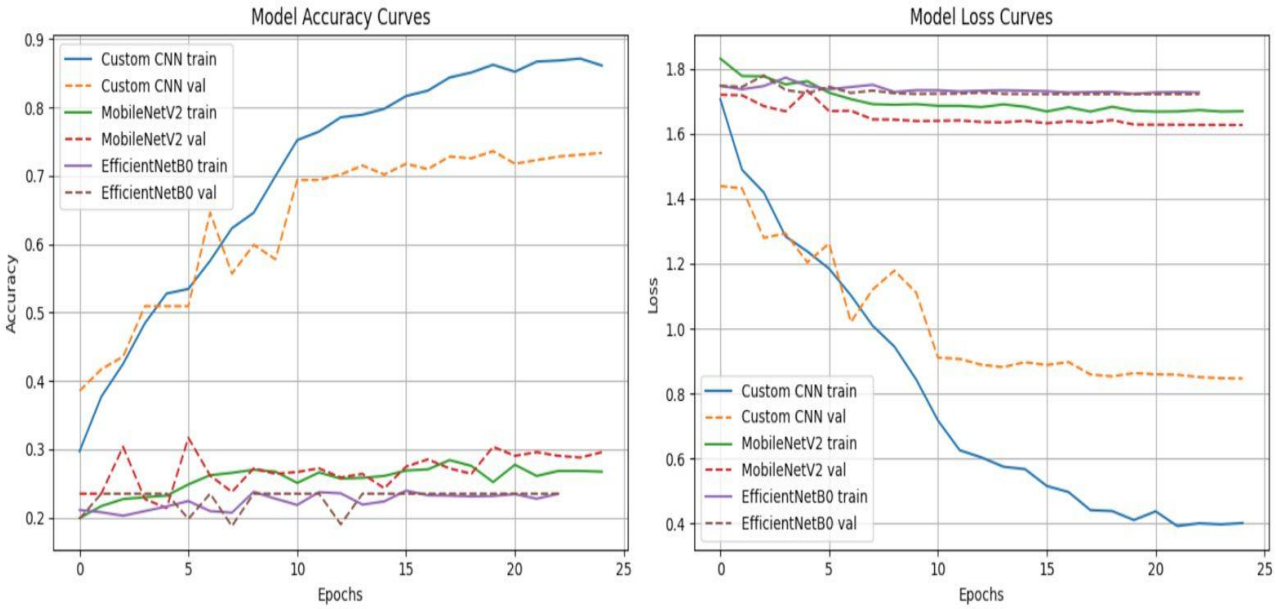


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FINAL EVALUATION OF BEST MODEL: Custom CNN
```

5. Analysis: Transfer Learning Performance:

Pre-trained models (MobileNetV2 and EfficientNetB0) performed poorly because ImageNet features do not align well with waste materials. Improvements require:

1. **Fine-tuning base layers:** Training base layers instead of freezing them.
2. **More training data:** Specifically, for the "trash" category.
3. **Better preprocessing:** Matching image normalization to pre-trained model expectations.
4. **Class weighting:** Addressing data imbalance during training.

6. Practical Workflow:

1. **Waste Detection:** IR sensor detects user presence.
2. **Image Capture:** Camera takes a photo of the waste item.
3. **Classification:** Custom CNN identifies the waste type.
4. **Fill Monitoring:** Ultrasonic sensor measures bin capacity; triggers 80% alert.
5. **Decision Making:** Random Forest predicts emptying urgency to optimize collection.

7. Recommendations:

1. **Use Custom CNN:** Currently the most effective for image-based classification (70% accuracy).
2. **Expand Dataset:** Add more images for "trash" and other underrepresented categories.
3. **Fine-tune Models:** Avoid freezing all layers in transfer learning architectures.
4. **Integrate Systems:** Use the Random Forest for logistics and the CNN for sorting.
5. **Additional Sensors:** Consider weight sensors for increased accuracy.

8. Future Work:

1. GPS tracking of bins
2. Real-time cloud dashboard
3. Solar-powered smart bins

9. Conclusion:

This project presents a hybrid smart waste management system that integrates IoT, machine learning, and deep learning techniques to improve the efficiency of waste collection and segregation. The sensor-based system, supported by a Random Forest model, provides reliable prediction of bin status with an accuracy of 89%, enabling optimized collection scheduling. Additionally, the image classification module using a custom CNN achieves 70% accuracy, demonstrating the feasibility of automated waste segregation into recyclable and non-recyclable categories.

Although the classification model requires further data and optimization for improved performance, the overall system successfully combines real-time monitoring, predictive analytics, and intelligent classification. This integrated approach reduces operational costs, minimizes human effort, and enhances urban cleanliness. The proposed system can be extended with cloud integration, mobile applications, and advanced AI models, making it a scalable solution for smart city waste management.

10. References:

1. TrashNet Dataset: Yang, M., & Thung, G. (2016). Classification of Trash for Recyclability Status. Stanford University. (Foundation for the waste image categories).

2. IoT Waste Monitoring: Anagnostopoulos, T., et al. (2017). Challenges and Opportunities of Waste Management in IoT-Enabled Smart Cities. (Supports the use of ultrasonic sensors and real-time alerts).
3. Random Forest Methodology: Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5-32. (Foundation for our bin status prediction model).
4. MobileNetV2 Architecture: Howard, A., et al. (2018). Inverted Residuals and Linear Bottlenecks. CVPR. (Explains the architecture we tested for transfer learning).
5. EfficientNet Methodology: Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. ICML. (Details on the second transfer learning model tested).
6. Arduino Documentation: Official Arduino Sensor Tutorials. (Used for implementing the ultrasonic and IR sensor averaging logic).